

DP-800 LAB 02



Connect to AdventureWorksLT

```
1  -- Verify key tables in AdventureWorksLT
2  SELECT TOP (5) CustomerID, FirstName, LastName
3  FROM SalesLT.Customer;
4
5  SELECT TOP (5) SalesOrderID, OrderDate, CustomerID
6  FROM SalesLT.SalesOrderHeader;
7
8  SELECT TOP (5) ProductID, Name, ListPrice
9  FROM SalesLT.Product;
10
```

100 % ✓ No se encontraron problemas.

Resultados Mensajes

	CustomerID	FirstName	LastName
1	1	Orlando	Gee
2	2	Keith	Harris
3	3	Donna	Carreras
4	4	Janet	Gates
5	5	Lucy	Harrington

	SalesOrderID	OrderDate	CustomerID
1	71774	2008-06-01 00:00:00.000	29847
2	71776	2008-06-01 00:00:00.000	30072
3	71780	2008-06-01 00:00:00.000	30113
4	71782	2008-06-01 00:00:00.000	29485
5	71783	2008-06-01 00:00:00.000	29957

	ProductID	Name	ListPrice
1	680	HL Road Frame - Black, 58	1431.50
2	706	HL Road Frame - Red, 58	1431.50
3	707	Sport-100 Helmet, Red	34.99
4	708	Sport-100 Helmet, Black	34.99
5	709	Mountain Bike Socks, M	9.50

Create a view to simplify queries

```
1  CREATE OR ALTER VIEW SalesLT.vCustomerOrders AS
2  SELECT
3      c.CustomerID,
4      CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,
5      h.SalesOrderID,
6      h.OrderDate
7  FROM SalesLT.Customer c
8  INNER JOIN SalesLT.SalesOrderHeader h ON c.CustomerID = h.CustomerID;
```

% ✓ No se encontraron problemas. Línea: 8, Ca

Mensajes

Los comandos se han completado correctamente.

Hora de finalización: 2026-09-21T11:52:38.2508273+02:00

```

1 SELECT TOP (5) *
2 FROM SalesLT.vCustomerOrders
3 ORDER BY OrderDate DESC;

```

100 % 2 0

Resultados Mensajes

	CustomerID	CustomerName	SalesOrderID	OrderDate
1	29847	David Hodgson	71774	2008-06-01 00:00:00.000
2	30072	Andrea Thomsen	71776	2008-06-01 00:00:00.000
3	30113	Raja Venugopal	71780	2008-06-01 00:00:00.000
4	29485	Catherine Abel	71782	2008-06-01 00:00:00.000
5	29957	Kevin Liu	71783	2008-06-01 00:00:00.000

Create a stored procedure to process an order

```

1 CREATE OR ALTER PROCEDURE dbo.AddOrderLineItem
2     @SalesOrderID INT,
3     @ProductID INT,
4     @Quantity INT
5 AS
6 BEGIN
7     SET NOCOUNT ON;
8     BEGIN TRANSACTION;
9
10    -- Use Product ListPrice as UnitPrice
11    DECLARE @UnitPrice DECIMAL(18,2);
12    SELECT @UnitPrice = CAST(ListPrice AS DECIMAL(18,2))
13    FROM SalesLT.Product
14    WHERE ProductID = @ProductID;
15
16    IF @UnitPrice IS NULL
17    BEGIN
18        ROLLBACK TRANSACTION;
19        THROW 50010, 'Invalid ProductID specified.', 1;
20    END
21
22    -- Ensure SalesOrderID exists
23    IF NOT EXISTS (SELECT 1 FROM SalesLT.SalesOrderHeader WHERE SalesOrderID = @SalesOrderID)
24    BEGIN
25        ROLLBACK TRANSACTION;
26        THROW 50011, 'Invalid SalesOrderID specified.', 1;
27    END
28
29    -- Insert line item (no discount)
30    INSERT INTO SalesLT.SalesOrderDetail (SalesOrderID, OrderQty, ProductID, UnitPrice, UnitPriceDiscount)
31    VALUES (@SalesOrderID, @Quantity, @ProductID, @UnitPrice, 0);
32
33    -- Update header subtotal based on current line totals
34    UPDATE h
35    SET SubTotal = d.SumLineTotal,
36        ModifiedDate = SYSUTCDATETIME()
37    FROM SalesLT.SalesOrderHeader h
38    INNER JOIN (
39        SELECT SalesOrderID, SUM(LineTotal) AS SumLineTotal
40        FROM SalesLT.SalesOrderDetail
41        WHERE SalesOrderID = @SalesOrderID
42        GROUP BY SalesOrderID
43    ) d ON d.SalesOrderID = h.SalesOrderID;
44
45    COMMIT TRANSACTION;
46 END;

```

Mensajes

Los comandos se han completado correctamente.

Hora de finalización: 2026-09-21T11:55:12.4535784+02:00

```

1  -- Add a line item to an existing order (choose a valid SalesOrderID)
2  DECLARE @SalesOrderID INT = (SELECT TOP 1 SalesOrderID
3                                FROM SalesLT.SalesOrderHeader
4                                ORDER BY SalesOrderID DESC);
5  EXEC dbo.AddOrderLineItem @SalesOrderID = @SalesOrderID,
6                                @ProductID = 680,
7                                @Quantity = 1; -- adjust ProductID as needed
8
9  SELECT TOP (5) *
10 FROM SalesLT.SalesOrderDetail
11 WHERE SalesOrderID = @SalesOrderID
12 ORDER BY SalesOrderDetailID DESC;
13
14 SELECT SalesOrderID, SubTotal, TaxAmt, Freight, TotalDue
15 FROM SalesLT.SalesOrderHeader
16 WHERE SalesOrderID = @SalesOrderID;|

```

	SalesOrderID	SalesOrderDetailID	OrderQty	ProductID	UnitPrice	UnitPriceDiscount	LineTotal	rowguid
1	71946	113407	1	680	1431.50	0.00	1431.500000	AEC11F49-B3D8-4B57-BE0E-B962B0
2	71946	113406	1	916	31.584	0.00	31.584000	B571CEE0-A90A-4E76-9796-379B55

	SalesOrderID	SubTotal	TaxAmt	Freight	TotalDue
1	71946	1463.084	3.1163	0.9738	1467.1741

Create a scalar function for reusable calculations

```

1  CREATE OR ALTER FUNCTION dbo.fnOrderTotal (@OrderID INT)
2  RETURNS DECIMAL(18,2)
3  AS
4  BEGIN
5      DECLARE @Total DECIMAL(18,2);
6
7      SELECT @Total = SUM(LineTotal)
8      FROM SalesLT.SalesOrderDetail
9      WHERE SalesOrderID = @OrderID;
10
11     RETURN ISNULL(@Total, 0.00);
12 END;|

```

100 %

✓ No se encontraron problemas.

Mensajes

Los comandos se han completado correctamente.

Hora de finalización: 2026-09-21T12:28:41.9930607+02:00

```

1 SELECT d.SalesOrderID, dbo.fnOrderTotal(d.SalesOrderID) AS OrderTotal
2 FROM SalesLT.SalesOrderDetail d
3 GROUP BY d.SalesOrderID
4 ORDER BY d.SalesOrderID DESC;

```

100 % 1 0 Línea: 4, Carácter: 31

	SalesOrderID	OrderTotal
1	71946	2894.58
2	71938	74160.23
3	71936	79589.62
4	71935	5533.87
5	71923	96.11
6	71920	2527.13
7	71917	37.76
8	71915	1732.89
9	71902	59894.21
10	71899	1856.21
11	71898	53248.69
12	71897	10585.05
13	71895	221.26
14	71885	524.66
15	71867	858.90
16	71863	2777.14
17	71858	11528.84
18	71856	500.30
19	71846	1884.39
20	71845	34118.54

Create an inline table-valued function (TVF)

```

1 CREATE OR ALTER FUNCTION dbo.GetCustomerOrders (@CustomerID INT)
2 RETURNS TABLE
3 AS
4 RETURN
5 (
6     SELECT
7         h.SalesOrderID,
8         h.OrderDate
9     FROM SalesLT.SalesOrderHeader h
10    WHERE h.CustomerID = @CustomerID
11 );

```

No se encontraron problemas.

Mensajes

Los comandos se han completado correctamente.

Hora de finalización: 2026-09-21T12:31:33.1475537+02:00

```

1 SELECT *
2 FROM dbo.GetCustomerOrders(29929)
3 ORDER BY OrderDate DESC;

```

100 % 2 0

	SalesOrderID	OrderDate
1	71902	2008-06-01 00:00:00.000

1	SELECT CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName, o.SalesOrderID, o.OrderDate
2	FROM SalesLT.Customer c
3	CROSS APPLY dbo.GetCustomerOrders(c.CustomerID) o
4	WHERE c.CustomerID = 29929;

100 %
3
0
Línea: 4, Carácter: 29
SPC
CRLF

Resultados
Mensajes

	CustomerName	SalesOrderID	OrderDate
1	Jeffrey Kurtz	71902	2008-06-01 00:00:00.000

Create a trigger to log changes

2	IF OBJECT_ID('dbo.OrderAudit') IS NULL
3	BEGIN
4	CREATE TABLE dbo.OrderAudit (
5	AuditID INT IDENTITY(1,1) PRIMARY KEY,
6	OrderID INT NOT NULL,
7	OldTotal DECIMAL(18,2) NULL,
8	NewTotal DECIMAL(18,2) NULL,
9	ChangedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME()
10	);
11	END
12	GO
13	
14	-- Trigger on order details updates
15	CREATE OR ALTER TRIGGER SalesLT.trg_LogOrderTotalChange
16	ON SalesLT.SalesOrderDetail
17	AFTER INSERT, UPDATE
18	AS
19	BEGIN
20	SET NOCOUNT ON;
21	
22	;WITH AffectedOrders AS (
23	SELECT SalesOrderID FROM inserted
24	UNION
25	SELECT SalesOrderID FROM deleted
26	),
27	-- New totals from the base table (already reflects changes)
28	NewTotals AS (
29	SELECT d.SalesOrderID, SUM(d.OrderQty * d.UnitPrice) AS Total
30	FROM SalesLT.SalesOrderDetail d
31	INNER JOIN AffectedOrders a ON d.SalesOrderID = a.SalesOrderID
32	GROUP BY d.SalesOrderID
33	),
34	-- Contribution of the newly inserted/updated rows
35	InsertedTotals AS (
36	SELECT SalesOrderID, SUM(OrderQty * UnitPrice) AS Total
37	FROM inserted
38	GROUP BY SalesOrderID
39	),
40	-- Contribution of the previous row versions (empty on INSERT)
41	DeletedTotals AS (
42	SELECT SalesOrderID, SUM(OrderQty * UnitPrice) AS Total
43	FROM deleted
44	GROUP BY SalesOrderID
45	)
46	INSERT INTO dbo.OrderAudit (OrderID, OldTotal, NewTotal)
47	SELECT
48	n.SalesOrderID,
49	n.Total - ISNULL(i.Total, 0) + ISNULL(d.Total, 0) AS OldTotal,
50	n.Total AS NewTotal
51	FROM NewTotals n
52	LEFT JOIN InsertedTotals i ON n.SalesOrderID = i.SalesOrderID
53	LEFT JOIN DeletedTotals d ON n.SalesOrderID = d.SalesOrderID;
54	END;

3 %
No se encontraron problemas.

Mensajes

Hora de finalización: 2026-09-21T12:36:24.9307534+02:00

SQLQuery12.s...25 (sa (80))*

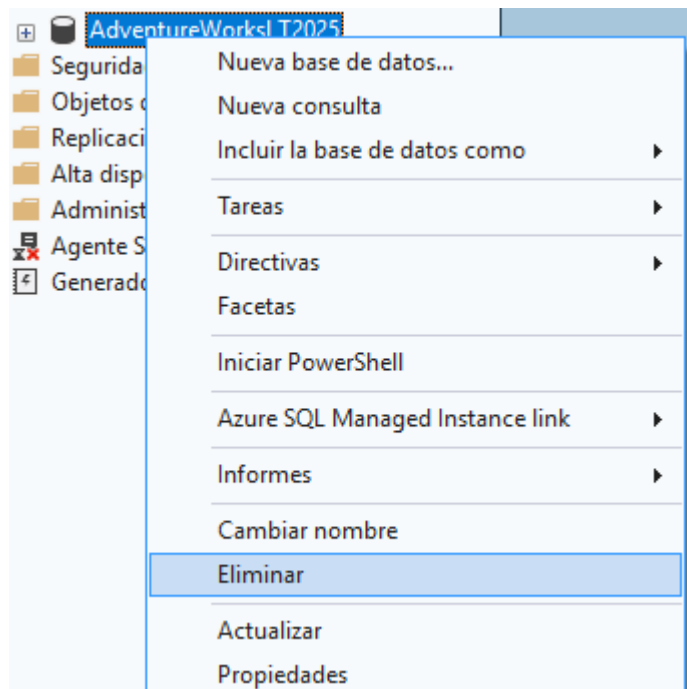
```
1  -- Update an order detail to change the total
2  UPDATE d
3  SET OrderQty = OrderQty + 1
4  FROM SalesLT.SalesOrderDetail d
5  WHERE d.SalesOrderID = (SELECT TOP 1 SalesOrderID FROM SalesLT.SalesOrderHeader ORDER BY SalesOrderID DESC);
6
7  SELECT TOP (5) *
8  FROM dbo.OrderAudit
9  ORDER BY AuditID DESC;
```

83 % 2 0 Línea: 9, Carácter: 24 SPC CRLF

Resultados Mensajes

	AuditID	OrderID	OldTotal	NewTotal	ChangedAt
1	1	71946	2894.58	5789.17	2026-09-21 10:37:49.7710359

Cleanup



Next steps

Enlace al repositorio de GitHub